

Assignment: Wine Quality Prediction Using ElasticNet Regression

Objective

The goal of this assignment is to predict the quality of wine based on its physicochemical properties using the **ElasticNet regression algorithm**. You will analyze the dataset, preprocess the data, build the regression model, and evaluate its performance.

Dataset Overview

The Wine Quality dataset contains physicochemical attributes of Portuguese "Vinho Verde" wine samples (both red and white), along with quality ratings. The objective is to model wine quality based on these physicochemical characteristics. The dataset is frequently used in regression and classification tasks in machine learning.

Dataset Source

- [UCI Machine Learning Repository - Wine Quality Dataset](#)
- Citation: "Modeling wine preferences by data mining from physicochemical properties," by P. Cortez, A. Cerdeira, Fernando Almeida, Telmo Matos, J. Reis. 2009 Published in Decision Support Systems

Columns in the CSV File

The dataset is presented as a CSV file with the following columns:

Column Name	Description
fixed acidity	Concentration of non-volatile acids (e.g., tartaric acid), expressed in g/dm ³ .
volatile acidity	Concentration of acetic acid, which can affect the taste negatively (g/dm ³).
citric acid	Concentration of citric acid, contributing to freshness and taste (g/dm ³).
residual sugar	Amount of sugar left after fermentation (g/dm ³).
chlorides	Salt content (g/dm ³).
free sulfur dioxide	SO ₂ in free form; prevents microbial growth and oxidation (mg/dm ³).
total sulfur dioxide	Total SO ₂ content, both free and bound forms (mg/dm ³).

Column Name	Description
density	Density of the wine; depends on alcohol and sugar (g/cm ³).
pH	Measure of acidity or basicity of the wine.
sulphates	Potassium sulphate level, contributing to flavor (g/dm ³).
alcohol	Alcohol content by volume (%).
quality	Target variable: Quality score of the wine (0 to 10, integer).

Key Notes

- **Data Characteristics:** All features are continuous and numeric, except for the `quality` column, which is an ordinal variable.
- **Use Cases:** This dataset is ideal for regression tasks (predicting `quality`) or classification tasks (grouping wine into quality categories).
- **Applications:** It has been used for predictive modeling in wine quality assessment and feature importance analysis.

The dataset provides a real-world scenario for exploring the relationships between physicochemical properties and wine quality.

Number of Attributes

- **Features (Input Variables):** 11 continuous numerical attributes
- **Target Variable (Output):** Wine quality (integer score from 0 to 10)

Problem Statement

You are tasked with developing a predictive model to estimate wine quality scores (target variable) based on a set of physicochemical attributes (input features). Your analysis should include data exploration, preprocessing, modeling, and evaluation steps.

Dataset

- Use the **Wine Quality Dataset** available from the [UCI Machine Learning Repository](#).
- Dataset URL: [Wine CSV](#)
- Hint: `read_csv` has special provisions to read datasets directly from URLs into Pandas DataFrames.
- Caution: while using `read_csv` with URLs, you need to make sure to specify the separator. So, for example, a valid usage would be

```
pd.read_csv(url, sep=';')
```

In [12]:

Print the first 10 rows of your DataFrame.

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Out[18]:

	fixed acidity	volatile acidity	citric acid	residual sugar	chlorides	free sulfur dioxide	total sulfur dioxide	density	pH	sulphates	quality
0	7.4	0.70	0.00	1.9	0.076	11.0	34.0	0.9978	3.51	0.56	
1	7.8	0.88	0.00	2.6	0.098	25.0	67.0	0.9968	3.20	0.68	
2	7.8	0.76	0.04	2.3	0.092	15.0	54.0	0.9970	3.26	0.65	
3	11.2	0.28	0.56	1.9	0.075	17.0	60.0	0.9980	3.16	0.58	
4	7.4	0.70	0.00	1.9	0.076	11.0	34.0	0.9978	3.51	0.56	
5	7.4	0.66	0.00	1.8	0.075	13.0	40.0	0.9978	3.51	0.56	
6	7.9	0.60	0.06	1.6	0.069	15.0	59.0	0.9964	3.30	0.46	
7	7.3	0.65	0.00	1.2	0.065	15.0	21.0	0.9946	3.39	0.47	
8	7.8	0.58	0.02	2.0	0.073	9.0	18.0	0.9968	3.36	0.57	
9	7.5	0.50	0.36	6.1	0.071	17.0	102.0	0.9978	3.35	0.80	

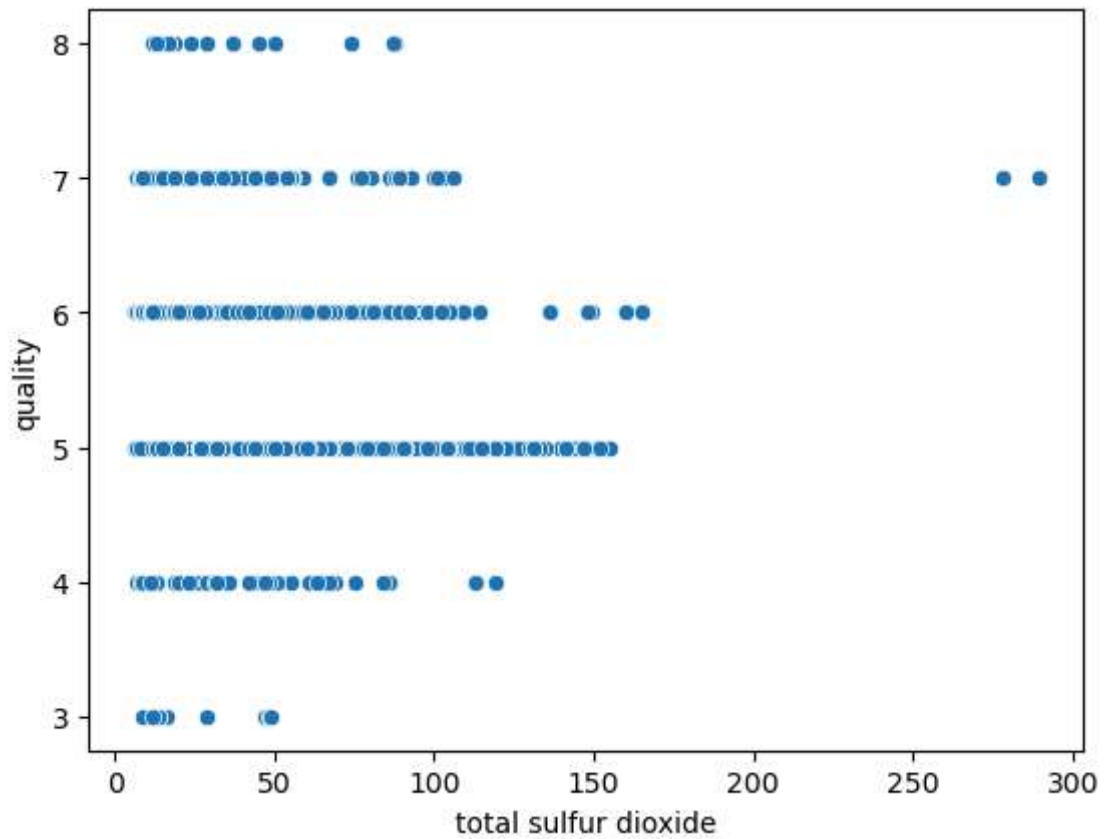
- Display summary statistics and check for missing values.
- What is the mean value of Citric Acid content in the dataset?
- How many missing values are there?
- What is the 3rd quartile of Total Sulfur Dioxide content?
- What is the mean quality of wines in the dataset?
- If `quality=0` describes the worst wine, what is the quality of the best wine in the dataset?

Data Visualization

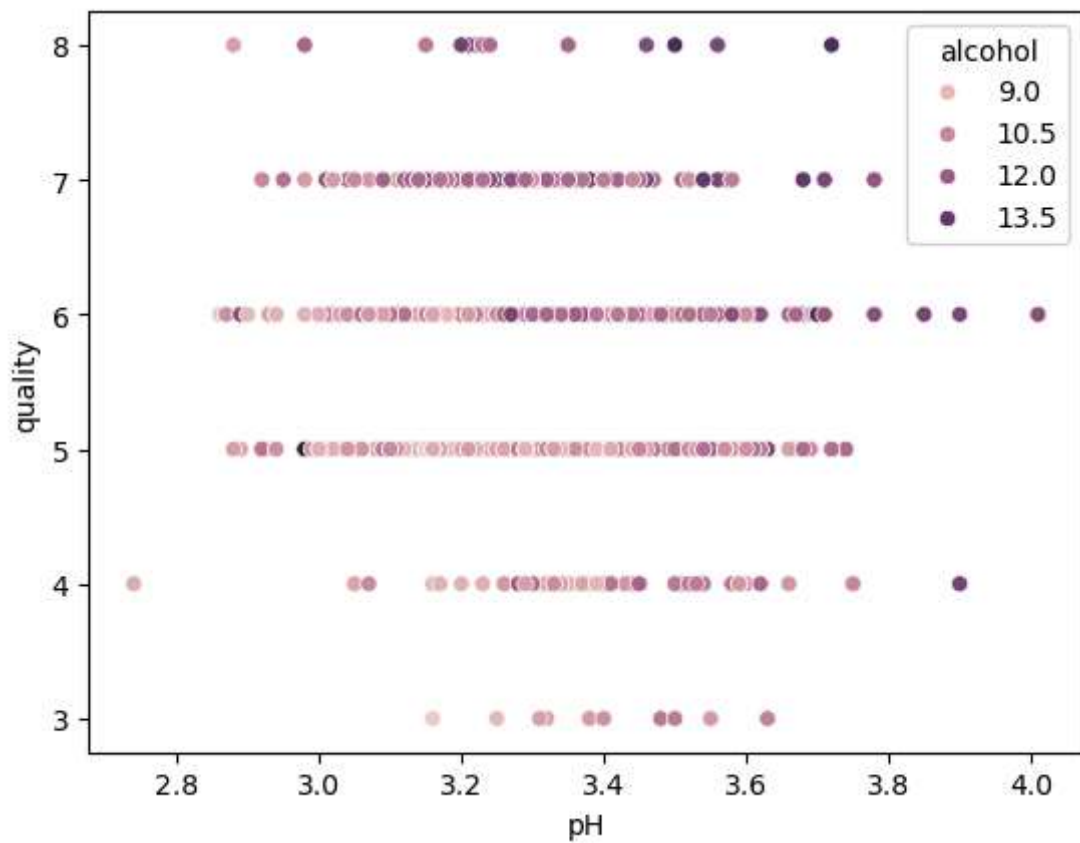
Visualize the relation between (a) wine quality and total sulfur dioxide, and (b) wine quality and pH as scatterplots. Remember to select the right `hue` .

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In [38]:



Data Preprocessing:

- Scale the features using the `StandardScaler` standardization technique.
- Split the data into training and testing sets (e.g., 80% training, 20% testing).

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Model Building:

- Implement the **ElasticNet regression** algorithm using Scikit-Learn.
- Explain in your own words what the `alpha` and `l1_ratio` parameters do. You will need to carefully examine the Scikit-Learn Documentation for `ElasticNet` to answer this question.
- [Link to ElasticNet documentation](#).
- Experiment with different values of `alpha` and `l1_ratio` to find the best combination for your model.
- Test your output using the following benchmark parameters: `alpha=1.0`, `l1_ratio=0.5`, `random_state=42`
- Mean Squared Error with the above parameters should be ≈ 0.657

In [46]:

Out[46]:

```

ElasticNet
ElasticNet(random_state=42)

```

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Hyperparameter Tuning

- Instead of `ElasticNet`, we will now use `ElasticNetCV` that also permits Cross Validation.
- Train an `ElasticNetCV` model on this dataset with the following parameters: `alphas=[0.01, 0.1, 1.0, 10.0]`, `l1_ratio=[0.1, 0.25, 0.5, 0.75, 0.9]`, and, as before, `random_state=42`.

In [64]:

Out[64]:

```

ElasticNetCV
ElasticNetCV(alphas=[0.01, 0.1, 1.0, 10.0],
              l1_ratio=[0.1, 0.25, 0.5, 0.75, 0.9], random_state=42)

```

- What are the best values of `alpha` and `l1_ratio`?

- Did using cross validation help in any way? Quantify the improvement or lack thereof.

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